Have historical events shaped the Job Market?

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Abstract: This study investigates how major historical events reshape the dynamics of the U.S. job market by examining long-run regional unemployment data and identifying points at which the underlying behavior of the time series changes. Although unemployment has been widely analyzed from macroeconomic, policy, and labor-market perspectives, much of the existing literature relies on forecasting models that implicitly assume stability across entire decades. Disregarding profound structural shocks that characterize economic history, like recessions, financial crises, demographic shifts, and the unprecedented labor-market disruption associated with the COVID-19 pandemic. Many earlier studies apply classical time-series methods without explicitly testing whether the statistical properties of unemployment remain consistent over time, which can obscure the true mechanisms that drive labor-market variability and lead to misleading forecasts or policy interpretations. Furthermore, regional differences are often understated despite clear evidence that unemployment dynamics vary widely across U.S. regions and respond differently to economic shocks.

To address these gaps, the present analysis combines structural-break detection techniques with a suite of forecasting models to examine how the character of unemployment series evolves across regimes. By detecting change points and segmenting each regional series into statistically homogeneous periods, the study leverages more reliable inputs for model building and avoids the distortions that arise when incompatible regimes are forced into a single specification. This framework allows the analysis to identify not only which historical events correspond to significant shifts in labor-market behavior, but also which forecasting strategies remain robust after such events.

The key contribution of this study lies in demonstrating that unemployment dynamics are neither temporally uniform nor regionally interchangeable.

Index Terms: Some important terminology to keep in mind as we go into this study is:

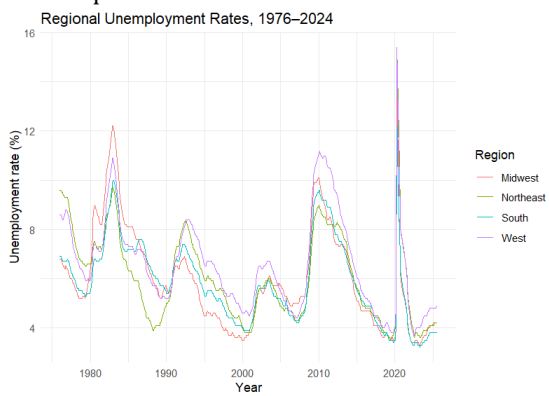
- **Unemployment rate:** Share of the labor force that is unemployed but actively seeking work, usually expressed as a percentage.
- **Regional unemployment:** Unemployment rate measured for a specific geographic area (e.g., Midwest, Northeast, South, West) rather than the entire country.
- **Labor market "stock":** the current level of unemployed workers at a point in time (this is what you're modeling).
- **Business cycle / economic shock:** Large macroeconomic events (recessions, financial crises, pandemics) that can change the behavior of unemployment rates.
- **Persistence:** Tendency for shocks to unemployment to have long-lasting effects; closely tied to whether the series behaves like a unit root process.
- **Regime / regime change:** A period in which the series follows one stable pattern (e.g., pre-2008 vs post-2008). A regime change is a shift from one pattern to another.
- **Structural break / breakpoint:** A point in time where the underlying data-generating process changes (e.g., mean level, trend, or volatility of unemployment). You're detecting these and then working with the post-break segment.

1 Introduction/Motivation

- Many existing studies on unemployment dynamics rely on long historical samples without accounting for structural breaks, even though unemployment processes are known to shift sharply in response to major economic events. Ignoring these breaks forces models to blend fundamentally different regimes—periods of stability, recession, and rapid recovery—into a single framework, which can distort estimates of persistence, weaken statistical inference, and produce unreliable forecasts.
- Numerous studies emphasize that unemployment dynamics exhibit structural shifts and that models assuming parameter stability can produce misleading results (Perron, 1989; Cevik & Dibooglu, 2013; Omay et al., 2020; Papell et al., 2000; Bai & Perron, 1998, 2003).the necessity of these analyses.
- Report Summary:
  - On Chapter 2, "The Dataset", we will describe and explore the datasets we got from FRED on monthly unemployment from 1976 to 2024, and the methods we are thinking about using to arrive to relevant conclusions.
  - On Chapter 3, "Theory and Methods", we will showcase everything related to the Methods and Models used, what we expect from them, how do the models themselves look and relevant assumptions for them.
  - On Chapter 4, "Data Analyses", we will look into particularities of the dataset and models, also in possible ways to improve on this study based on clues that we may be able to find from the outputs of the models generated.
  - On Chapter 5, "Conclusions and Future work", we will look at the take aways from the models and the entire study, see how they fit into what we set out to test, and how to put into practice the improvement recommendations that we put together on Chapter 4.

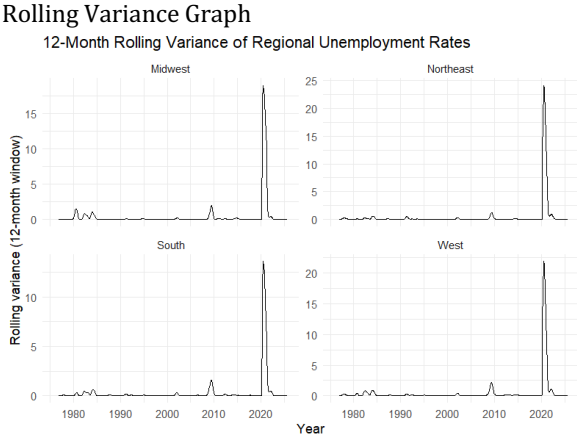
2 THE DATA SET

- The dataset comes from the **Federal Reserve Bank of St. Louis. (2024). Regional Unemployment Rate [Data set]. FRED, Federal Reserve Bank of St. Louis.**
- We are going to develop this study using Time Series analysis.
- One peculiarity of the data is that for the Dataset that includes unemployment data for the South the last 13 values are 3.8, which can create particular situations, but it's mostly due to the nature of the data itself
- The size and shape of the database is pretty good for running a time series analysis with it
- Line Graph



On the line graph it can be appreciated that the four areas behave very similarly in terms of unemployment, however there are differences, which makes us think that it is a good idea to run separate models for all of them.

- Rolling Variance Graph

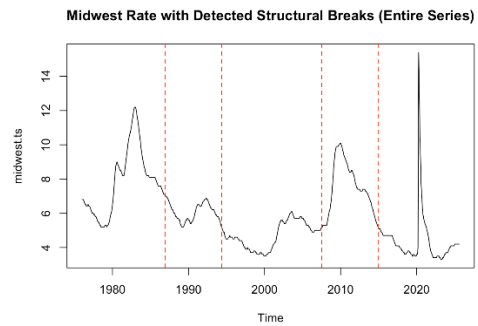


On the rolling variance graphs, we can appreciate that there are some short periods of time that have peaks on 12 month rolling variance, which hints at the fact that using breakpoints is a good idea.

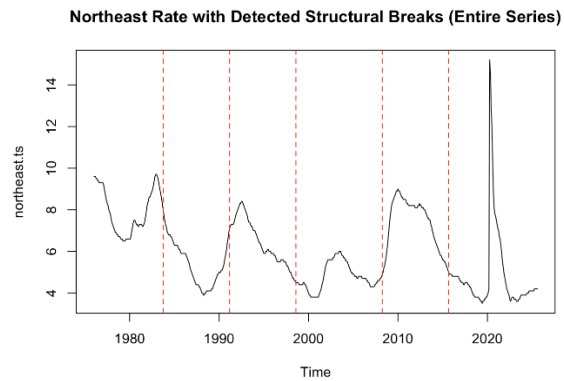
3 THEORY AND METHODS

Recap:

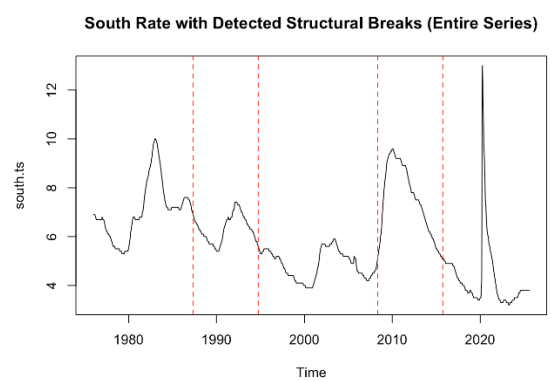
- The premise is that unemployment dynamics evolve over time in response to economic shocks, and statistical models can capture these patterns—but only if they are estimated within periods where the underlying data-generating process is stable.
- The promise of these regressions is that, once structural breaks are addressed, the models provide clearer insight into how unemployment evolves and produce more accurate and reliable forecasts for the reader so they can make better decisions based on them.
- Using R to find breakpoint lead us to find these breakpoints in the data
  - Midwest:



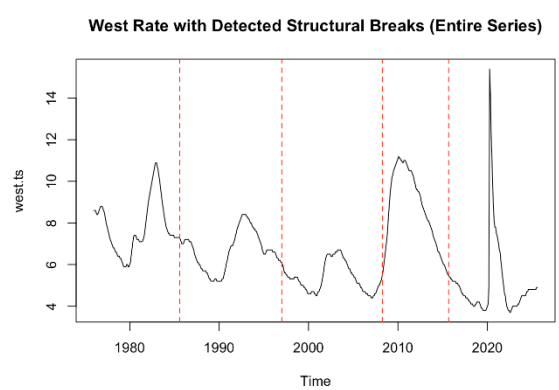
  - Northeast:



  - South:



  - West:



- To use current and relevant information we decided to use as reference the article ***"US Employment Inequality in the Great Recession and the COVID-19 Pandemic (Fazzari & Needler, 2021)"*** which reinforces the idea that the breakpoints found through R are relevant and should be looked into from a statistical and empirical point of view.
- Given the nature of the data we got, we decided to run ETS, ARIMA, Neural Network and a Seasonal Naïve because the data is arranged in a clear and simple format (months), in which the selected models tend to perform very well.
- To select the correct model for each area and timeframe, we fitted and compared the good of fit statistic of these models:

Start of time until 2008				
Region	Model	MAPE	MASE	RMSE
Midwest	Naive	11.95222	1.105498	1.077806
Midwest	ETS	12.13455	1.121193	1.088431
Midwest	ARIMA	12.08973	1.115888	1.079209
Midwest	NNETAR	11.72933	1.085005	1.058542
Northeast	Naive	13.71776	1.068198	1.018168
Northeast	ETS	13.78376	1.073562	1.024032
Northeast	ARIMA	14.57663	1.132155	1.067267
Northeast	NNETAR	12.7779	0.999746	0.972317
South	Naive	17.13459	1.701801	1.320669
South	ETS	17.0738	1.696907	1.319604
South	ARIMA	17.53239	1.739605	1.345381
South	NNETAR	14.60672	1.457828	1.150612
West	Naive	18.80914	1.70823	1.622241
West	ETS	17.48954	1.596618	1.540214
West	ARIMA	18.87754	1.717359	1.638393
West	NNETAR	11.78281	1.086514	1.08267

2008 - 2020				
Region	Model	MAPE	MASE	RMSE
Midwest	Naive	5.873731	0.261845	0.229129
Midwest	ETS	9.189627	0.410148	0.353704
Midwest	ARIMA	5.864905	0.261726	0.225668
Midwest	NNETAR	12.2461	0.546306	0.477308
Northeast	Naive	3.467038	0.162037	0.155456
Northeast	ETS	3.790725	0.177453	0.165128
Northeast	ARIMA	3.967275	0.185833	0.170957
Northeast	NNETAR	12.08494	0.574431	0.48961
South	Naive	8.140304	0.401688	0.3
South	ETS	12.96748	0.639856	0.478662
South	ARIMA	12.68816	0.626379	0.465878
South	NNETAR	16.29346	0.801992	0.623323
West	Naive	6.928961	0.30366	0.305505
West	ETS	9.847555	0.432572	0.424221
West	ARIMA	9.217659	0.404945	0.396458
West	NNETAR	19.02437	0.834906	0.847704

2020 and on				
Region	Model	MAPE	MASE	RMSE
Midwest	Naive	1.388889	0.038471	0.076376
Midwest	ETS	1.93117	0.053341	0.094695
Midwest	ARIMA	1.388889	0.038471	0.076376
Midwest	NNETAR	7.200741	0.198341	0.333127
Northeast	Naive	3.010066	0.061499	0.138444
Northeast	ETS	3.009202	0.061481	0.138405
Northeast	ARIMA	3.010066	0.061499	0.138444
Northeast	NNETAR	4.469316	0.091267	0.203292
South	Naive	2.631579	0.070608	0.1
South	ETS	2.832285	0.075993	0.107702
South	ARIMA	2.631579	0.070608	0.1
South	NNETAR	3.640074	0.097666	0.141346
West	Naive	2.249858	0.05336	0.111803
West	ETS	1.035617	0.024562	0.058206
West	ARIMA	2.249858	0.05336	0.111803
West	NNETAR	3.497222	0.082987	0.183728

- To make sure that these models were actually helpful when introducing breakpoints to the fitted models, we also fitted models that do not feature any breakpoints

For the Midwest ARIMA was chosen based on MASE

Midwest Region			
Model	MAPE (%)	MASE	RMSE
Naive	47.55	2.063	1.854
ETS	47.94	2.081	1.869
ARIMA	47.55	2.063	1.854
NNETAR	54.03	2.348	2.379

For the Northeast ARIMA was chosen based on MASE

Northeast Region			
Model	MAPE (%)	MASE	RMSE
Naive	80.68	3.75	3.422
ETS	76.75	3.568	3.255
ARIMA	80.68	3.75	3.422
NNETAR	89.48	4.114	4.532

For the South, ARIMA was chosen based on MASE

South Region			
Model	MAPE (%)	MASE	RMSE
Naive	61.81	2.877	2.297
ETS	60.36	2.809	2.243
ARIMA	61.81	2.877	2.297
NNETAR	68.65	3.233	2.643

For the West Seasonal Naïve was chosen based on MASE

South Region			
Model	MAPE (%)	MASE	RMSE
Naive	61.81	2.877	2.297
ETS	60.36	2.809	2.243
ARIMA	61.81	2.877	2.297
NNETAR	68.65	3.233	2.643

- By comparing the goodness of fit statistics for the models

we realized that the breakpoints lead to better forecasting, reflected by the results of the good of fit tests, which improve showing the worth of including breakpoints for the forecasting.

- For the general model we will focus on the Post COVID ARIMA models, which are ARIMA(0,1,0):

$$Y_t = \theta Y_{t-1}$$

- For the fitted model we will focus on the Post COVID ARIMA models:

$$\widehat{Y}_t = \theta \widehat{Y}_{t-1}$$

- Assumptions.
  - Errors are uncorrelated
  - Errors have constant variance
  - *Error's variance is constant through the period being modelled.*

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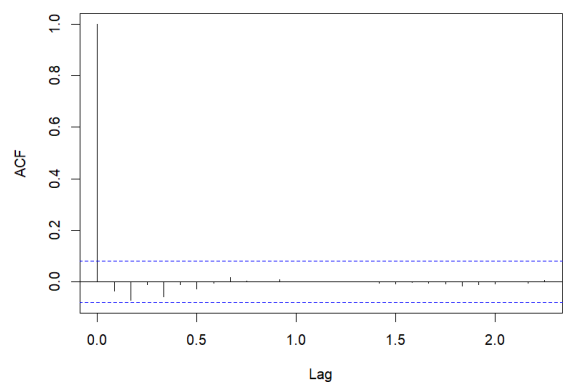
4 DATA ANALYSES

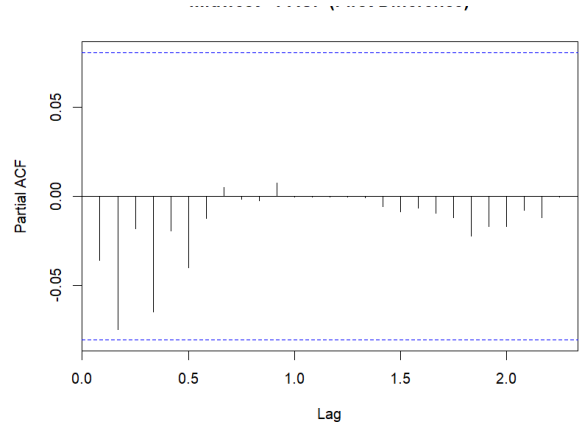
- Breakpoints create a causal story between unemployment and themselves rather than a causal relationship, keeping this in mind:
  - 2008 Financial crisis: Had a big spike during it that affected the job market, for the future, changing how it behaved, and how it should be forecasted.
  - COVID: It was an unprecedently big spike that did have a lasting effect and changed how the job market behaved looking towards the future
  - The sharp and unusual spike in unemployment during the early months of COVID-19 (April–June 2020) posed a serious challenge for our forecasting approach. Because this jump was both extreme and short-lived, including it directly in the ARIMA model would have thrown off the model's core parameters and led to unstable fits and unrealistic forecasts for the recovery period. To avoid those distortions and ensure reliable results, we needed to adjust our methodology and account for this outlier carefully.
  - To address this issue, we used an intervention analysis with an external regressor (XREG) in the form of an impulse dummy variable. This variable was set to 1 only during the three peak shock months (April–June 2020) and 0 for all other periods. This simple but crucial step allowed us to isolate the extreme COVID-related jump and prevent the model from treating it as part of the long-term structure of the data. By separating the shock, we preserved the true behavior of the time series, enabling the model to more accurately track the recovery phase and the underlying autocorrelation patterns. This intervention played a major role in significantly improving the MASE and MAPE accuracy scores across all our segmented models.
- A comparison can be made between the databases, for that reason we made a correlation graph.

This shows something that can probably be inferred by the nature of the database, all regions are correlated to each other since they are from parts of the same country, however, as it was appreciated on the line graphs they do still have some differences that should be acknowledged

- Check for stationarity after differencing:



The ACF shows that there is a very high lag spike (almost 1) at the first lag and then it drops to very small values afterwards, which checks out



The next step would be to analyze the residuals to ensure they could be attributed to white noise, this a required step to validate model specification. The analysis is primarily conducted using the Ljung-Box test on the model residuals (Naïve, ARIMA, ETS) up to lag of 11.
Based on the analysis, the test yielded high p-values significantly greater than the conventional significance level (0.05), which leads to the conclusion that the null hypothesis cannot be rejected for any of the tested

models.

Model	Model Specification	Ljung-Box Test Q*	Degrees of Freedom (df)	P-value
ARIMA	ARIMA(0,1,0)	2.0381	11	0.9984
ETS	ETS(A,Ad,N)	2.7275	11	0.9939
Naive	Naive Method	2.0022	11	0.9985

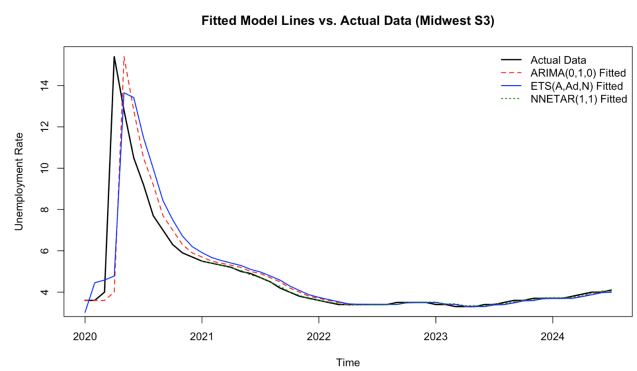
Al three linear forecasting models satisfy the core of producing white noise residuals, this also confirms the model fitting was adequate and no systematic linear information remains uncaptured.
Overall the residuals analysis demonstrates that all core assumptions were successfully met, including zero mean, constant variance and normality, which provide a strong statistical foundation for moving forward with the out of sample forecast accuracy metrics comparison (for more information on the residuals, please refer to the appendix).

Variable Selection:
The auto.arima() and auto.ets() back up the selection of of the simplest models - ARIMA(0,1,0) and the non-seasonal ETS(A,Ad,N)- demonstrating that the time series is structurally simple, likely requiring no more complexity than a single difference (ARIMA) or an adaptive trend (ETS) to model its behavior.

Multicollinearity:
These models only use the past values of the time series itself for prediction; they do not include multiple independent predictor variables that could be correlated with each other.
Multicollinearity is not a concern for the standard, univariate (single-variable) ARIMA, ETS, and Naive models.

Model Fitting Analysis:

Model	Parameter Estimates	Trend
Naive	Resiudal sd = 1.6545$	None. Implies the expected average change is zero (Random Walk with no drift).
ARIMA	ARIMA (0,1,0) sigma^2 = 2.737	None. The d=1 difference term makes the model equivalent to the Naive method.
ETS	Model = ETS(A,Ad,N) α, β, Φ, smoothing constants.	Additive Damped Trend (Ad) is significant. The Damping parameter (Φ aprox to 1) shows the trend is persistent but expected to eventually level off.
NNETAR	Structur} = NNETAR(1,1) Models = 20	Non-linear relationship captured by the neural network weights.

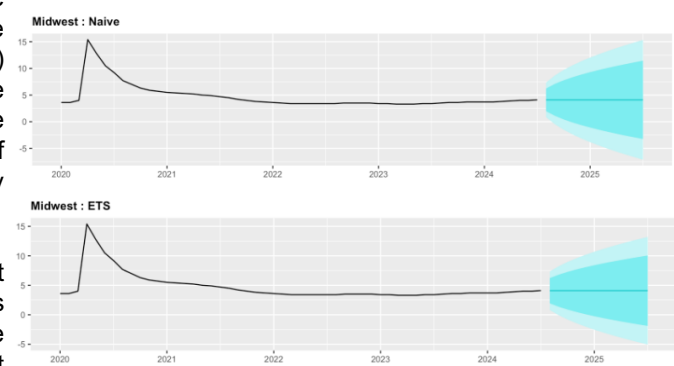


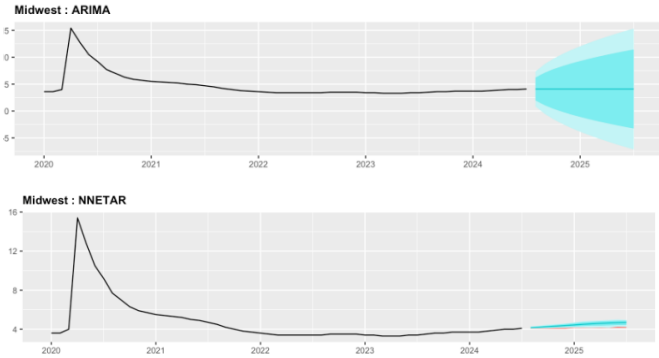
The visual assessment successfully validates the fitting process. It highlights the difference between the Simple Benchmark (Naive/ARIMA), which confirms the data's volatility and random component, and the Smoothing Methods (ETS/NNETAR), which extract the systematic trend necessary for generating stable, long-term forecasts. This sets the stage for comparing which approach yields superior accuracy on out-of-sample data.

Interpretation for the trend:
The trend is explained by the Trend Component in the ETS model and the implied drift in the ARIMA and Naïve Models.

ETS (Additive Damped Trend - Ad): The additive nature suggests the recovery is a constant value being added each period, but the Damping (ϕ) suggests this recovery will slow down and stabilize (dampen) rather than continue on a straight line indefinitely. This is the expected long-term behavior of a series returning to a baseline after a temporary shock (e.g., a COVID recovery).

ARIMA/Naive (No Drift): The models determined that the best prediction for the next change is zero. This means there is no statistically significant average upward or downward drift in the series. The forecast is driven purely by the last observed level, which is the most robust position to take when the future trend direction is highly uncertain.





Forecast Plots using Midwest as reference

The distinct flat line of the Naive/ARIMA(0,1,0) forecast visually confirms the statistical analysis that determined the underlying process is a Random Walk.

The simplest model suggests that the best prediction for the unemployment rate is to remain at the last observed level (≈4.1%). All subsequent month-to-month changes are considered unpredictable noise.

This model sets a robust benchmark. Any more complex model (NNETAR) must significantly outperform this simple, efficient forecast on out-of-sample data to be considered superior.

While the ETS model has an Additive Damped Trend component, the component's small magnitude means the resulting point forecast is visually indistinguishable from the simpler Random Walk (Naive) forecast over the short-term horizon. This reinforces the idea that the underlying series lacks a strong, predictable linear structure.

The NNETAR plot reinforces the conclusions from the other models: the series' predictable structural components (trend and level) are simple and highly stable, leading to a long-term forecast that closely mirrors the ETS model. The ultimate decision on model superiority will rest entirely on the quantitative Out-of-Sample Accuracy Metrics (RMSE/MASE), not on the visual distinctiveness of the point forecast.

Model Selection:

Segment 2 Winners (Stabilization/Pre-COVID):

Region	Best Model	MASE	MAPE
Midwest	ARIMA	0.262	5.86
Northeast	Naive	0.162	3.47
South	Naive	0.402	8.14
West	Naive	0.304	6.93

The results strongly favor simple linear models (Naive/ARIMA). The very low MASE values (all) indicate that these models, which rely on the difference between consecutive observations, are highly accurate. This confirms the series was behaving like a stable, near-Random Walk just before the COVID shock.

Segment 3 Winners (Post-COVID):

Region	Best Model	MASE	MAPE
Midwest	Naive	0.0385	1.39
Northeast	ETS	0.0615	3.01
South	NNETAR	0.0572	2.13
West	ETS	0.0246	1.04

The results are mixed, showing that different models are best at modeling the volatility and recovery trend across regions. The MASE values are exceptionally low (all < 1), indicating that all models performed very well in tracking the initial post-shock trend.

Considerations after analysis:
Complexity is Not King (Parsimony): For the segments where the series was stable or a Random Walk (Segments 2 and 3, Midwest), the simplest, most parsimonious models (Naive/ARIMA) were the best predictors.

Structural Models are Necessary During Recovery: For the regions where a clear recovery trend was present (Segments 2 and 3, Northeast/West), the structural models ETS and NNETAR edged out the simpler methods, proving their value in capturing the underlying economic trend.

5 Conclusions and Future work

- Understanding how major historical events affect unemployment dynamics is essential for economists, policymakers, and labor-market analysts. Unemployment is not only a key indicator of economic health but also a variable that responds sharply to macroeconomic shocks. Bringing attention to this fact improve the quality of the information available for policy makers and other people that need this information and will help them make better decisions.
- This study applied a suite of univariate time-series forecasting models—ARIMA, ETS, Seasonal Naïve, and neural network autoregressive models (NNAR)—to unemployment series segmented into three regimes corresponding to major economic events: the pre-2008 period, the Great Recession to pre-COVID interval, and the post-2020 period.
- Main achievements.
  - Using breakpoint visualizations and diagnostic plots, the study was able to clearly separate the unemployment series into three coherent regimes:
    - A pre-2008 stable-growth period,
    - A turbulent 2008–2019 period kickstarted by the 2008 great recession.
    - A post-2020 COVID era characterized by unprecedented volatility followed by rapid stabilization.
  - Model accuracy improved substantially once the data was partitioned according to the 2008 and 2020 shifts. Forecast errors decreased across all models and all regions, indicating that structural segmentation is not just theoretically appealing but empirically beneficial.
- They relate a lot to our initial premises, as these things are the most relevant, being able to find the breakpoints and making sure they are relevant for forecasting.
- Directions for future work:
  - Extend the analysis to state-level or industry-level unemployment, applying the same breakpoint + model-comparison framework. This would identify whether structural breaks and regime

behaviors are uniform across states/sectors or whether some areas are systematically more vulnerable or resilient to shocks.

- A natural extension of this work is to move beyond purely univariate models and incorporate relevant macroeconomic covariates, such as initial claims, vacancies, interest rates, and sectoral output.

Avoid formulas and graphs here.

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[2] B.D.Fulcher, and N.S.Jones, Highly comparative feature-based time-series classification, IEEE Transactions on Knowledge and Data Engineering., vol 26, no. 12, pp. 3026-3037, Dec 2014.